CS_5080_Lecture_12

Thursday, March 5, 2026 4:36 PM

Neural Networks

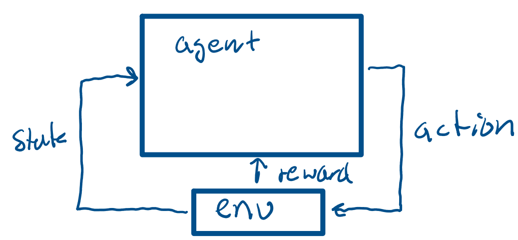
State input → Q(s,a) table → output
deocrete
functions

function approximation
Convo neural net
generalized format of
Q(s,a) table

deep Q-learning – throw multiple
states into the CNN or
Q (s,a) table. Has a history/
memory of prior actions

ANNs in RL

Generic policy learning neural net



Neurons in the brain

100 Billion neurons – connected to 3-4000 other neurons

Dendrite – fibers connected to soma
Synapse – connecting point between axons
Soma – body of neuron
Nucleus – center of cell
Axon – connection to other neurons
Axonal arborization

Natural Nueral Nets

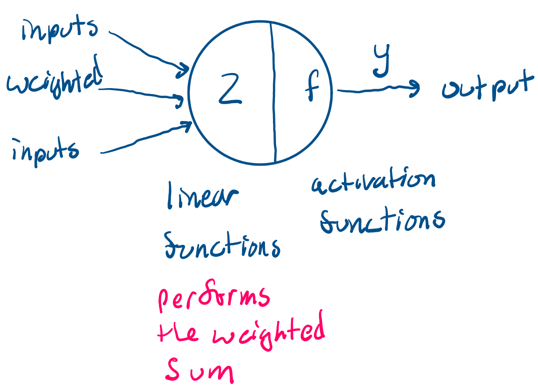
Signals - electro chem process

Synapse - communicate via chemical transmitters
can be inhibitory or excitatory

ANN

McCullogh and Pitts 1943
generialzed designers of first ANN

Modeling the neron



Characterizing an ANN

- Activation function at a single neuron level - step, sign, sigmoid....
- Architecture at network level - organized / connected
- Learning algor at network level - connections weight changed / learned over time

Examples of Activation functions
- slide has a large table of functions
  Claude.ai - see if you can find this in the book
  ReLU - Hinton - Toronto

Learning - Setting the weights
- Supervised training - labeled data
- Un supervised - has to figure out everything on its own

Rules:

Back propagation - see AI slides from last semester

Loss function - again see slides from last semester

Epoch of training - hopefully it learns something every epoch

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Architecture : • Single layer Feed forward ANN or Perception

laurene fausett , 1990

classification of letters

• Multilayer Neural Networks

in layer
hidden layer(s)
output layer

universal func approximator? Claude.ai, look this term up
one hidden layer
not practical to compute

Receptive Fields of vision
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